



Baseline assessment of biochemical biomarkers in different tissues of blue crab (*Callinectes sapidus*)



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INTRODUCTION

ATLANTIC BLUE CRAB

- ❖ Significantly impacting biodiversity and fisheries in estuarine Mediterranean habitats: omnivorous, opportunistic and versatile predator
- ❖ Emerging as human seafood source in invaded habitats
- ❖ Opportunity as “sentinel” species to monitor marine pollution
- ❖ Contaminant-related biomarkers and health effects scarcely investigated

BIOCHEMICAL BIOMARKERS

Early indicators for the diagnosis of environmental risk

- ❖ **Phase I biotransformation:** Cytochrome P450 (P450) NAD(P)H-cytochrome P450 reductase (P450-reductase), EROD activity
- ❖ **Phase II biotransformation:** glutathione S-transferase (GST)
- ❖ **Metal detoxification:** Metallothionein (MT)
- ❖ **Neurotoxicity:** Acetylcholinesterase (AChE)
- ❖ **Oxidative stress:** superoxide dismutase (SOD), catalase (CAT) glutathione peroxidase (GP), glutathione reductase (GR)
- ❖ **Oxidative damage:** Lipid peroxidation (LPO)



SAMPLING

April 2024. Mar Menor lagoon

LONG-TERM HUSBANDRY

Individual confinement, closed filtered water system, 70 days, 42-44 ‰, 20-22 °C, natural daylength, environmental enrichment, fed every two days frozen seafood

SUBCELLULAR FRACTIONS

- 10.000g- cytosol (AChE)
- 30.000g- cytosol (MT)
- 36.000g- cytosol (Phase II, oxidative stress)
- 36.000g microsomes (Phase I and LPO)

BIOMETRY

Sex, Carapace length and width (CL, CW)
Body weight (BW), Gonadosomatic index (GSI), Hepatosomatic index (HSI), Condition Factor (CF)

OBJECTIVES

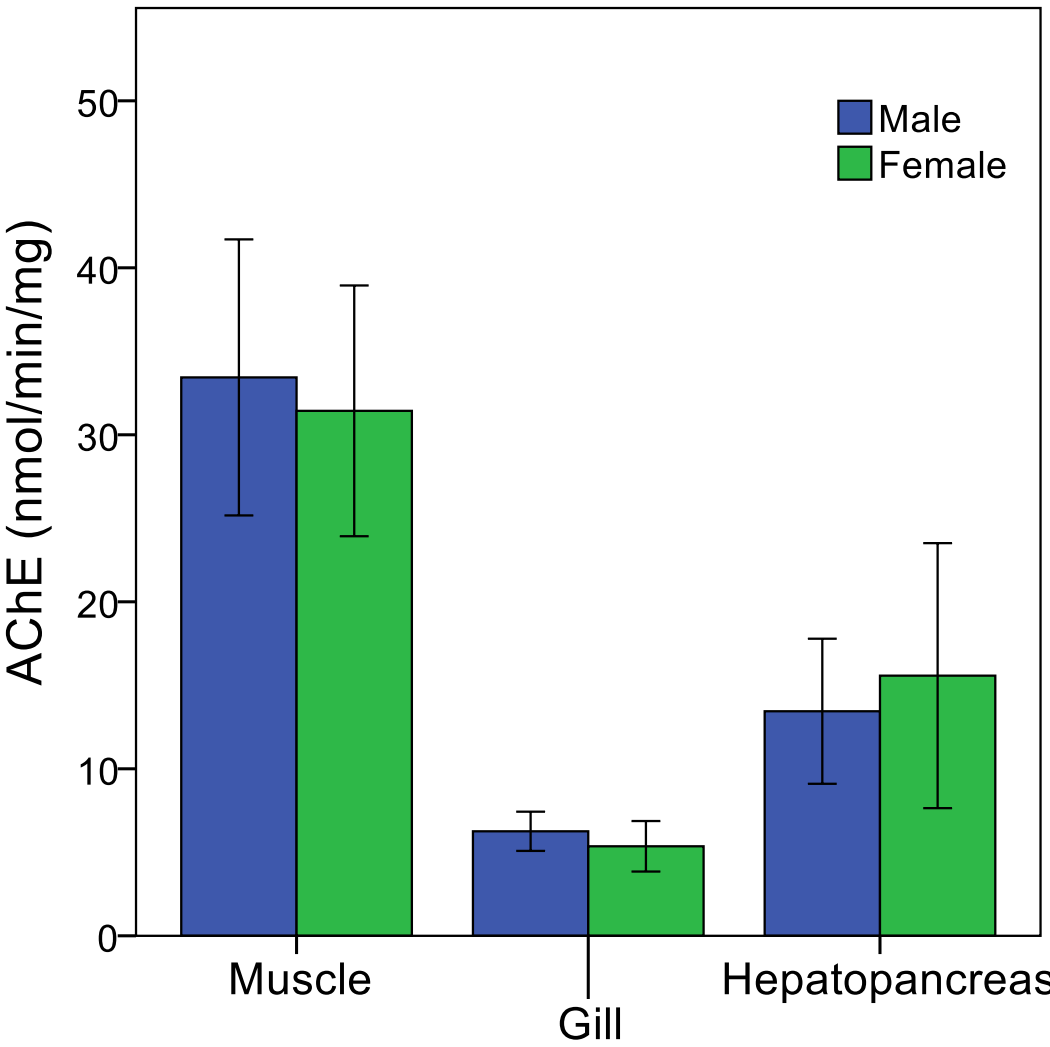
- ❖ Long-term captive husbandry of blue crabs to facilitate field and laboratory research.
- ❖ Validation of ecotoxicological biomarkers for environmental monitoring applications.
- ❖ Identification of optimal tissues and subcellular fractions for biomarker quantification.
- ❖ Determination of baseline biomarker responses in different tissues under standardized conditions

Biometry	Sex	CL (mm)	CW (mm)	BW (g)	GSI	HSI	CF
FEMALE	4	4	4	4	2	4	4
mean		61	126	112	0.5	3	89
sd		5	11	19	0.2	1	8
MALE	12	12	12	12	11	12	12
mean		61	123	137	0.8	3	111
sd		4	9	28	0.4	2	16
TOTAL	16	16	16	16	13	16	16
mean		61	123	130	0.8	3	105
sd		4	9	28	0.4	2	17

RESULTS AND DISCUSION

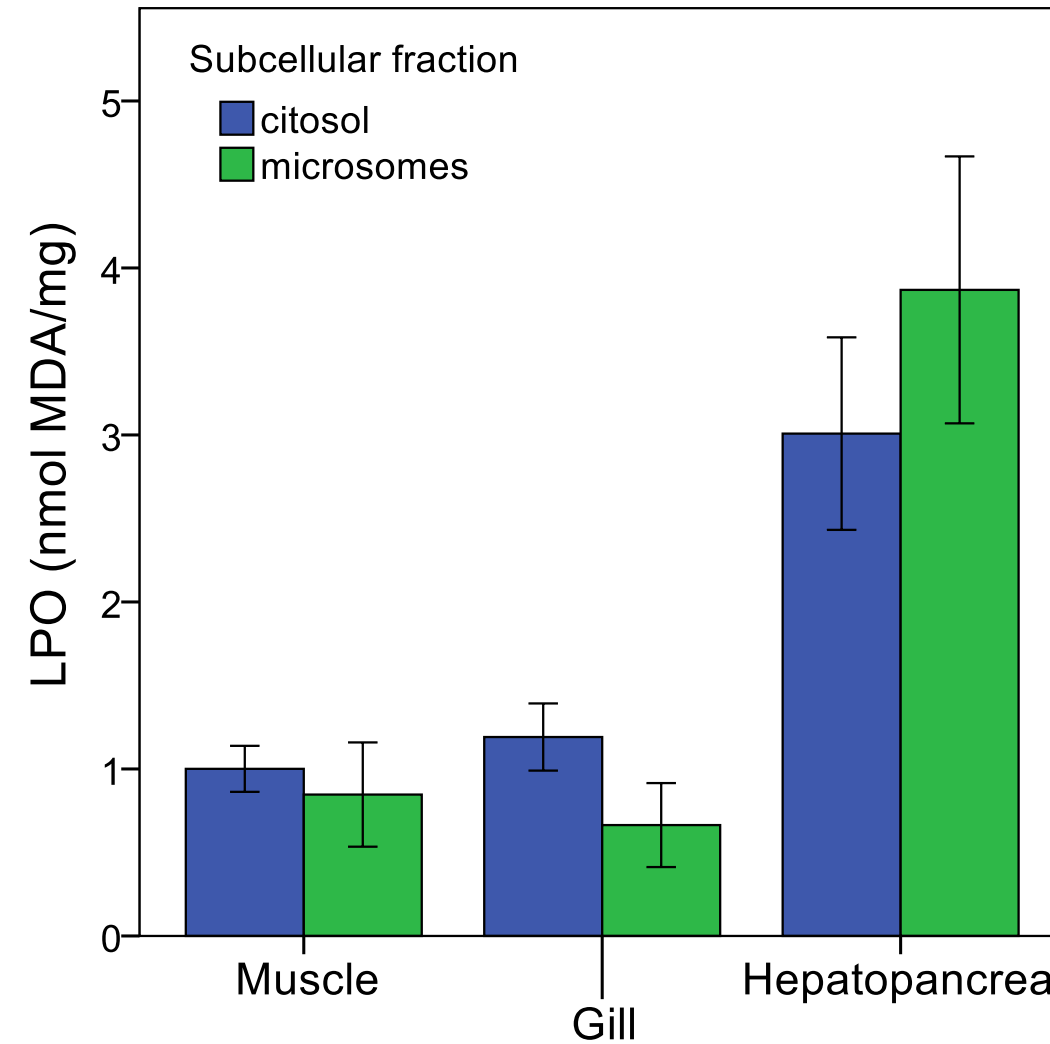
			AChE		GST		SOD		CAT		GST		GR	
			(nmol/min/mg)		(nmol/min/mg)		(Units/mg)		(μmol/min/mg)		(nmol/min/mg)		(nmol/min/mg)	
Tissue		n	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD	mean	SD
MUSCLE	Total	16	32.9	7.9	1.7	0.6	2.6	1.0	6.4	2.0	2.2	1.3	1.76	0.59
	Male	12	33.4	8.3	1.7	0.7	2.4	1.0	6.6	2.0	2.0	1.4	1.64	0.41
	Femal	4	31.4	7.5	1.7	0.3	3.1	0.5	6.0	2.2	2.6	0.8	2.13	0.94
GILL	Total	16	6.0	1.3	13.4	4.0	5.7	1.8	27.3	9.6	9.5	2.1	2.54	1.05
	Male	12	6.3	1.2	14.5	4.0	5.6	1.8	24.2	8.3	9.3	2.3	2.57	1.21
	Femal	4	5.4	1.5	10.2	2.5	6.2	1.8	36.7	7.7	10.2	1.3	2.46	0.41
HEPATOPANCREAS	Total	16	14.0	5.2	16.6	7.6	8.1	3.8	71.3	88.5	17.3	6.0	1.28	0.82
	Male	12	13.4	4.3	18.5	7.7	7.8	4.1	78.4	102.1	16.1	6.3	1.24	0.88
	Femal	4	15.6	7.9	11.0	4.0	9.0	3.2	51.7	33.0	21.0	3.6	1.39	0.69
HAEMOLYMPH	Total	16	0.11	0.04	0.07	0.04	4.7	2.2	na		na		na	
	Male	12	0.11	0.04	0.08	0.04	4.7	2.6	na		na		na	
	Femal	4	0.11	0.02	0.06	0.02	4.7	1.0	na		na		na	

na: non analysed; Tissue with greater response of each biomarker highlighted in green



Acetylcholinesterase

- ❖ No differences between sexes
- ❖ Highest levels in muscle
- ❖ Muscle appeared to be the most suitable tissue for AChE analysis in the blue crab



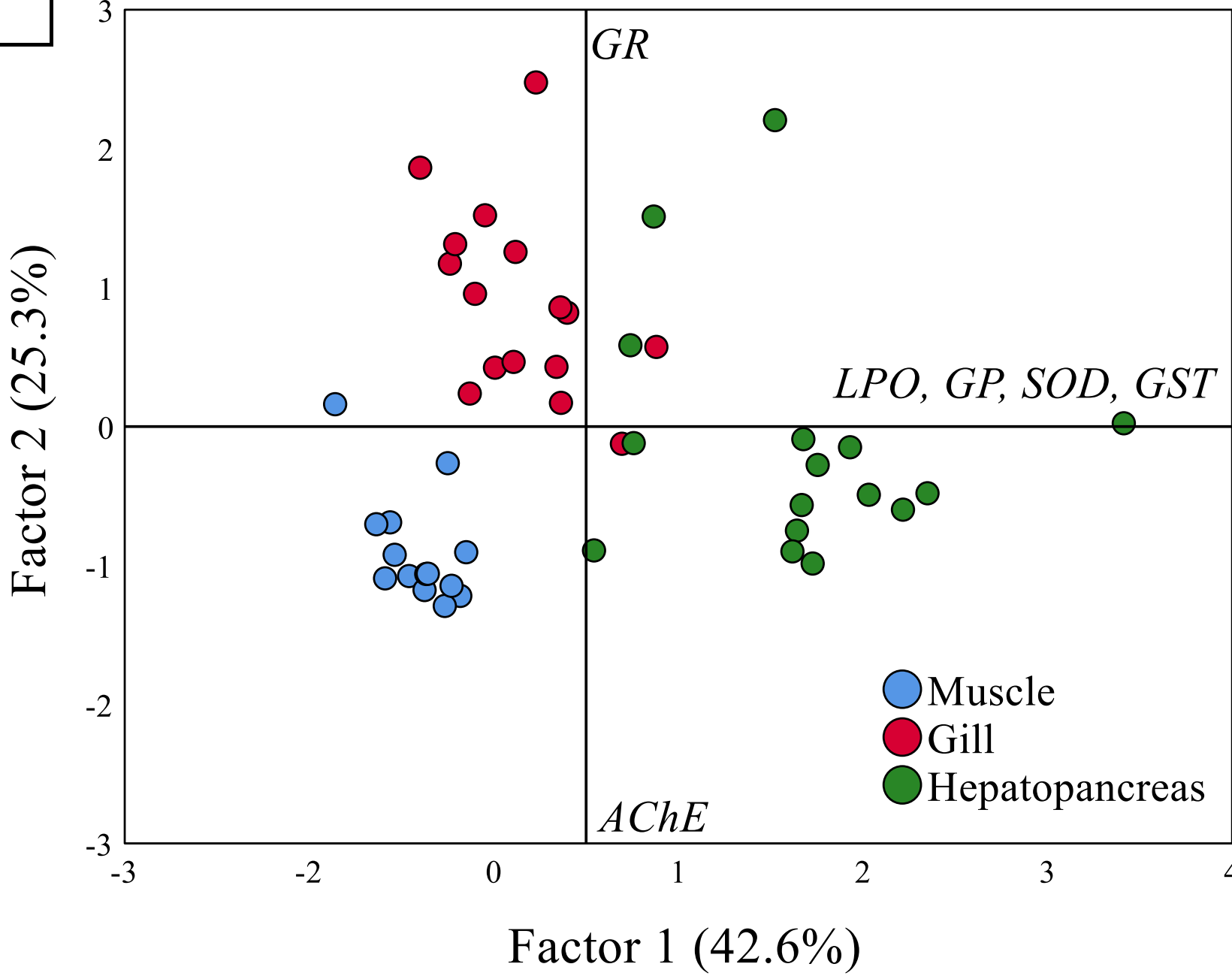
Lipid peroxidation

- ❖ Highest levels in hepatopancreas
- ❖ Comparable levels in the cytosolic and microsomal fractions of the hepatopancreas
- ❖ Hepatopancreas appeared to be the most suitable tissue for LPO analysis in the blue crab

❖ P450, P450-reductase and EROD activities were non detected

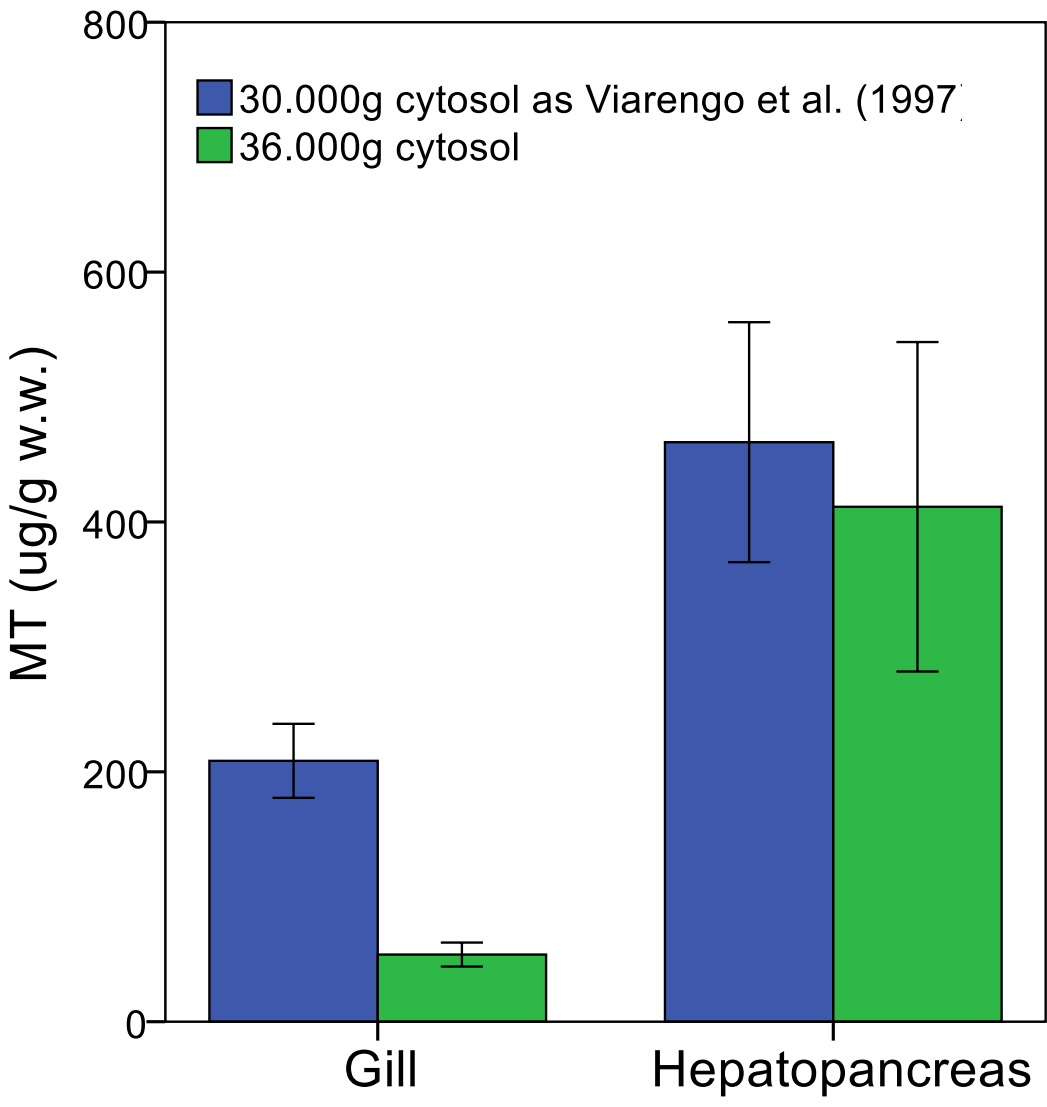
❖ The measurement of Phase I detoxification components in the hepatopancreas presents a methodological challenge due to the degradation of the cytochrome P450 complex during microsomal isolation

❖ Haemolymph showed the lowest levels of AChE and GST, but SOD levels comparable to those of other tissues



Principal Component Analysis

- ❖ Gill showed the highest level of GR
- ❖ Muscle showed the highest level for AChE
- ❖ Hepatopancreas showed the highest responsiveness for LPO, GP, SOD, GST
- ❖ Despite its greater responsiveness, the hepatopancreas also showed the greatest inter-individual variability, which means a lower ability to draw reliable conclusions when a limited number of samples are available.



Metallothioneins

- ❖ Highest levels in hepatopancreas.
- ❖ Subcellular fractions obtained using the original procedure of Viarengo et al. (1997) showed better recoveries in gill but similar ones in hepatopancreas.
- ❖ The limited information available on MT and metals in the blue crab supports measuring MT in both tissues until a decision is made on which tissue is best to be used for biomonitoring